Manav Mepani

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EDUCATION

Worcester Polytechnic Institute (WPI), USA **Aug 2024 – May 2026**
Master of Science in Robotics, **CGPA – 4.0/4.0**

Sardar Patel College of Engineering (SPCE), India **Aug 2018 – Jun 2022**
Bachelor of Technology in Mechanical Engineering, **CGPA – 9.04/10**

EXPERIENCE

EssilorLuxottica – GENTEX (MA, USA) | *Automation Support Intern* **May 2025 - Present**

- Improved production quality and reduced mispack risk by deploying automated vision-based inspection checks, ensuring label correctness and traceability before shipment.
- Troubleshot and programmed robotic arms and automated production-line equipment, resolving mechanical issues (grippers, alignment, sensors) and software-related faults to minimize downtime.

  Key skills: *Preventive and Corrective Maintenance, Root Cause Analysis in Manufacturing Environments, Documentation & Technical Writing*

The Innovation Story (INDIA) | *Mechanical Design Engineer (Robotics)* **Aug 2023 – Apr 2024**

- Designed and manufactured complex systems() for game challenges, achieving ~90% reliability within three months using innovative designs and in-house manufacturing.
- Worked with teams of 20 students of various grades while teaching them physics, CAD, and manufacturing for the competition in which they stood 4th all over India.

  Key Skills: *CNC, Lathe, 3D printing, CAD Designing, Design for manufacturing, Teamwork, Python Programming, Mechatronics*

Technology Innovation Hub, IIT Bombay (INDIA) | *Project Research Assistant* **Jun 2022 – Aug 2023**

- Designed and manufactured an autonomous onion harvester() for small farms, featuring an Energy-Aware Path Planning algorithm and a return-to-base function triggered by low battery levels to ensure operational efficiency.
- Improved the maneuverability of a bomb disposal rover by using a design that ensures at least one side's wheels maintain ground contact for enhanced stability using dynamic analysis software ADAMS.
- Optimized rocker-bogie mechanism design for a rover using Simulated Annealing optimization algorithm in Octave to enhance step-climbing ability by analyzing key design parameter relationships.

  Key Skills: *3D printing, CAD Designing, Design for manufacturing, FEA, Dynamic Analysis, Teamwork, Problem solving, optimization algorithm*

Prolific 3D Tech (INDIA) | *Intern Design Engineer* **Jun 2021 - Jul 2021**

- Led a reverse engineering project to replicate and modify a Can Filling and Seaming Machine(), creating 3D models and 2D manufacturing drawings in SOLIDWORKS after precise part measurements.

SOFTWARE AND TECHNICAL COMPETENCIES

Softwares: CATIA V5/SOLIDWORKS, AutoCAD, ANSYS Workbench, ADAMS, FluidSIM, MS OFFICE SUIT, Studio 5000 (Allen-Bradley)

Programming Skills: Arduino, Python, Octave, Robot Operating System (ROS2), MATLAB, Epson RC+ (SPEL+)

Certifications(): Neural Networks and Deep learning, Python Programming for data structure and accessing web data, Digital Electronic circuits, ANSYS Workbench, Siemens PLC

PUBLICATIONS

Energy-and Safety-Aware Operation of Battery-Powered Autonomous Robots in Agriculture()
IEEE Transactions on AgriFood Electronics (2024).

Study of the Step Climbing Ability of a Rover with Rocker Bogie Mechanism()
Journal of The Institution of Engineers (India): Series C (2024): 1-17.

Design of Robot Arm for Domestic Culinary Assistance()
Materials Today: Proceedings, Volume 68, Part 6, 2022, Pages 1930-1945, (ICAME22 Conference)

ACADEMIC PROJECTS

Undergrad Thesis: Design and Development of an Autonomous Cooking System() **Aug 2021 – May 2022**

- Designed and fabricated a 6 DOF **PUMA-configured** robotic arm for autonomous cooking.
- Performed **static structural** and **thermal analysis** for a 0.6 Kg payload and heat from cooking in ANSYS, verified through testing.
- Simulated **Inverse Kinematics algorithm** which used **kinematic decoupling method** in **Webots** and controlled the physical model using an Arduino microcontroller.
- Achieved a **0.7 Kg (1.55 lbs) payload capacity** with moderately high repeatability of **±1° error** in motion control using PID.